

Manas Joshi

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Honors Computing Science student specializing in real-time Machine Learning (ML) systems and backend development. Possessing demonstrable experience in constructing low-latency inference pipelines, RESTful APIs, and ML applications integrated with hardware. Enthusiastically seeking Software Engineering (SWE) or AI-focused co-op opportunities.

TECHNICAL SKILLS

Languages: Python, C/C++, Java, RISC V Assembly, JavaScript

Backend & Systems: Flask, Django, REST APIs, Linux, Asynchronous Programming

AI/ML: PyTorch, TensorFlow, Scikit-learn

Databases & Tools: MySQL, Git, Docker, UNIX, Bash, Shell Scripting

Core CS: Data Structures, Algorithms, OOP, Complexity Analysis

EDUCATION

University of Alberta

BSc (Honors) Computing Science

Relevant Coursework: Machine Learning, Reinforcement Learning, Algorithms, Computer Architecture

Edmonton, AB

Expected May 2027

EXPERIENCE

Bionix · EMG & RL Systems (Design Team)

Signal Processing Team Member

Edmonton, AB

Oct 2025 – Present

- Developed and rigorously tested novel Reinforcement Learning (RL) control algorithms for an EMG-driven prosthetic leg, optimizing responsiveness, gait stability, and user functionality.
- Executed EMG data acquisition, meticulous labeling, and pre-processing. Validated developed RL control policies using the standardized RL-Glue framework.
- Fostered interdisciplinary collaboration (EE, CS, and BME), accelerating project milestones and driving innovation in assistive robotics.

PROJECTS

NeuroMotion | Python, PyTorch, Flask, JavaScript, HTML, CSS

Nov 2025

- Pioneered a real-time EEG classification pipeline, achieving ultra-low latency (<200 ms) for immediate data processing and analysis.
- Designed and deployed a scalable RESTful Flask backend to deliver real-time, high-availability model inference.
- Engineered an efficient system utilizing sliding-window signal processing and asynchronous prediction handling for optimized performance.
- Seamlessly integrated OpenBCI hardware, establishing a robust and scalable data ingestion infrastructure.
- Secured 2nd place in the NatHacks competition by achieving a high-fidelity validation accuracy of 87%.

SkyNav | Python, Scikit-learn, Flask, JavaScript, HTML, CSS

Feb 2025

- Implemented A* pathfinding with dynamic weather constraints, reducing route calculation time by 15%.
- Designed a scalable backend supporting real-time route recomputation for over 10,000 concurrent users, improving reliability by 25%.
- Integrated OpenWeather API with asynchronous handling, reducing data retrieval latency by 20% and ensuring 99.9% uptime for weather-based routing.
- Developed high-performance REST endpoints for live UI updates, achieving sub-200ms latency for route changes.

Math-Vision-Model | Python, TensorFlow, Matplotlib

Jul 2024 – Jan 2025

- Developed and optimized a ResNet-based image classification pipeline, achieving a high-performance 88% accuracy across 89 distinct classes.
- Established a reproducible machine learning workflow, implementing rigorous version control for model exports (.h5) to ensure traceability and deployment reliability.
- Streamlined the data ingestion process by automating dataset preprocessing, integrating a real-world dataset (EMNIST) with synthetically generated data to enhance training diversity.
- Significantly improved model stability and generalization through systematic hyperparameter tuning and the design of a structured, comprehensive evaluation framework.

natHACKS 2023 – Video Game | Unity, C#, C++

Nov 2023

- Successfully developed and implemented advanced reinforcement learning control strategies for effective EMG-driven prosthetic assistance.
- Designed and built robust signal preprocessing and real-time EMG feature extraction pipelines, enhancing system responsiveness.
- Rigorously evaluated policy effectiveness and performance using the industry-standard RL-Glue experimentation framework.
- Played a key role in facilitating critical cross-functional collaboration with Electrical Engineering, Computer Science, and Biomedical teams, driving successful assistive robotics integration.

INTERESTS AND ACTIVITIES

- Electric Guitar
- Badminton
- Swimming
- Tennis